How Accurate Must a Free-Return Translunar Burn Be, and What Does Correcting an Injection Error Cost?

AE 370 Group Project 1
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1 Premise

1.1 Motivation

A *free-return trajectory* carries a spacecraft from Earth, around the Moon, and back to Earth without any further major propulsive maneuver. If the main engine fails after translunar injection (TLI), the Moon's gravity alone bends the trajectory home, which makes free return the default abort-safe design for crewed lunar missions.

Apollo 13 shows both why free return matters and why it is not sufficient on its own. A midcourse correction on the second day had moved the spacecraft off free return onto a *hybrid* trajectory to reach the Fra Mauro landing site, so when the oxygen tank failed the spacecraft was not on a path that would return it to Earth [1]. At 61 h 30 min mission elapsed time (MET), the crew fired the Lunar Module descent engine for 34 s to re-establish free return [1]. Further corrections followed after the lunar flyby, including a 14.8 s burn at 105 h 18 min MET to fine-tune the reentry path [1]. Mission Control described the acceptable entry corridor as approximately $-5.25°$ to $-7.4°$ of flight-path angle at entry interface, and targeted the final correction to $-6.52°$ [2].

Returning to Earth's vicinity is therefore not enough: the spacecraft must reach the atmosphere within a narrow band of *entry flight-path angle* γ_{EI}. Too shallow, and the capsule skips off the atmosphere; too steep, and deceleration and heating exceed crew and vehicle limits. Modern capsules such as Orion define their entry corridor in terms of γ_{EI} and the vehicle's lift-to-drag ratio [3]; on Artemis I, Orion reached entry interface at $\gamma_{\mathrm{EI}} = -5.66°$ and an inertial speed of 36,063 ft/s (≈ 10.99 km/s) [4], inside the Apollo corridor.

The lunar flyby strongly amplifies small errors in the TLI burn. Simulating our nominal free return (defined below) shows that, to first order, a 1 mm/s error in TLI magnitude changes γ_{EI} by about $0.14°$, and a 1 s error in burn timing changes it by about $5.7°$. The entire $2.15°$ corridor therefore corresponds to only about 15 mm/s of TLI magnitude and about 0.4 s of burn timing: accuracy at the millimeter-per-second and sub-second level, beyond what an engine can deliver open-loop,

which is why midcourse corrections exist. This project quantifies those accuracy requirements and the propellant cost of correcting an imperfect burn.

1.2 Research questions

Q1. Injection accuracy. How sensitive is the entry flight-path angle γ_{EI} to errors in (a) TLI burn magnitude and (b) TLI burn timing? What are the largest errors in each that keep γ_{EI} within the corridor $-7.4^\circ \le \gamma_{\mathrm{EI}} \le -5.25^\circ$ [2]?

Q2. Correction cost. When a TLI error places the spacecraft outside the corridor, how large an impulsive midcourse correction Δv_c is required to restore $\gamma_{\mathrm{EI}} = -6.52^\circ$, and how does that cost depend on the correction time t_c? In particular, is it cheaper to correct before or after the lunar flyby, and by how much?

Hypothesis. We expect the allowable TLI errors to be on the order of millimeters per second and fractions of a second, and the correction cost to depend strongly on t_c, with the lunar flyby marking a clear transition. The numerical study is designed to confirm or refute both.

1.3 Model and assumptions

We model the spacecraft with the *planar circular restricted three-body problem* (CR3BP) [6], under the following assumptions:

1. Earth and Moon are point masses on circular orbits about their common barycenter.
2. The spacecraft's mass is negligible and does not perturb Earth or Moon.
3. Motion is confined to the Earth–Moon orbital plane; the Moon's orbital inclination is neglected.
4. Solar gravity, solar radiation pressure, and Earth oblateness (J_2) are neglected.
5. All burns are impulsive (instantaneous velocity changes).
6. There is no atmosphere. γ_{EI} is evaluated where the vacuum trajectory crosses the entry-interface altitude, consistent with its use as a targeting criterion in mission design.

1.4 The initial value problem

Figure 1 shows the geometry: the spacecraft departs a low Earth parking orbit, swings behind the Moon, and returns to cross the entry interface, where the entry flight-path angle γ_{EI} is measured.

Nondimensionalization. Lengths, times, and velocities are scaled as in Table 1. Earth lies at $(-\mu, 0)$ and the Moon at $(1-\mu, 0)$ in a frame rotating with the Earth–Moon line at unit angular rate.

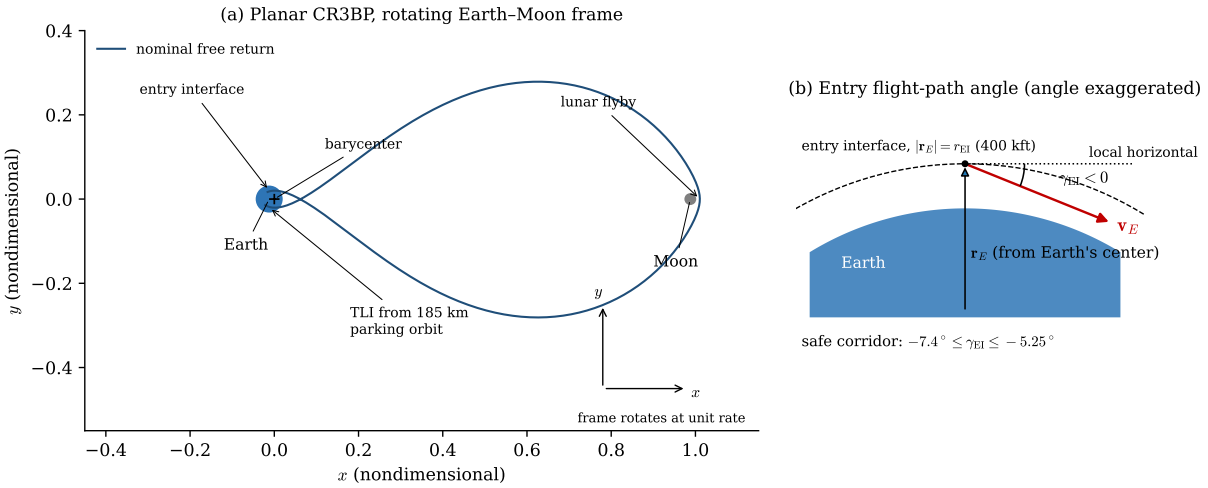


Figure 1: (a) The nominal free return in the rotating Earth–Moon frame, computed with our RK4 implementation (Earth and Moon not to scale). (b) Definition of the entry flight-path angle γ_{EI} at entry interface: the angle between the Earth-relative inertial velocity $\mathbf{v}_E$ and the local horizontal, negative when descending.

Table 1: Nondimensional units and problem constants.

Quantity	Symbol	Dimensional	Nondimensional
Length unit (Earth–Moon distance)	L	384,400 km	1
Time unit (inverse orbital rate)	T	4.3425 days	1
Velocity unit	$V = L/T$	1.0246 km/s	1
Mass parameter	$\mu = m_M/(m_E + m_M)$	—	0.012154
Earth radius	R_E	6,378.165 km	0.016593
Moon radius	R_M	1,737.4 km	0.004520
Parking orbit radius (185 km alt.)	r_0	6,563.2 km	0.017074
Entry interface radius (400,000 ft alt.)	r_{EI}	6,500.1 km	0.016910

The time unit is not set independently: the nondimensional equations assume $G(m_E + m_M) = L^3/T^2$, so $T = \sqrt{L^3/G(m_E + m_M)}$ with $m_E = 5.9722 \times 10^{24}\,\mathrm{kg}$ and $m_M = 7.3477 \times 10^{22}\,\mathrm{kg}$. The entry-interface altitude of 400,000 ft follows the convention used for Orion [4].

Governing equations. With state $\mathbf{u} = [x,\ y,\ v_x,\ v_y]^T$ in the rotating frame,

$$\begin{aligned} \dot{x} &= v_x, \\ \dot{y} &= v_y, \\ \dot{v}_x &= 2v_y + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3}, \\ \dot{v}_y &= -2v_x + y - \frac{(1-\mu)\,y}{r_1^3} - \frac{\mu\,y}{r_2^3}, \end{aligned} \tag{1}$$

where

$$r_1 = \sqrt{(x+\mu)^2 + y^2}, \qquad r_2 = \sqrt{(x-1+\mu)^2 + y^2} \tag{2}$$

are the distances to Earth and Moon. The $\pm 2v$ terms are the Coriolis acceleration and the bare x, y terms the centrifugal acceleration. Equation (1) is a coupled, nonlinear, autonomous IVP,

$$\dot{\mathbf{u}} = \mathbf{f}(\mathbf{u}), \qquad \mathbf{u}(0) = \mathbf{u}_0. \tag{3}$$

Initial condition: TLI from a parking orbit. The spacecraft starts on a circular, prograde 185 km parking orbit at angle φ from the Earth–Moon line and performs a tangential TLI burn of magnitude Δv (dimensional, km/s):

$$\begin{aligned} x_0 &= -\mu + r_0 \cos\varphi, \qquad & y_0 &= r_0 \sin\varphi, \\ v_c &= \sqrt{\frac{1-\mu}{r_0}} + \frac{\Delta v}{V}, & & \\ v_{x,0} &= -v_c \sin\varphi + y_0, \qquad & v_{y,0} &= v_c \cos\varphi - (x_0 + \mu). \end{aligned} \tag{4}$$

The final line converts the inertial velocity to the rotating frame via $\mathbf{v}_{\mathrm{rot}} = \mathbf{v}_{\mathrm{in}} - \boldsymbol{\omega} \times \mathbf{r}$, including Earth's motion about the barycenter. Burn *timing* maps to φ: the parking orbit turns at $n_{\mathrm{LEO}} = \sqrt{(1-\mu)/r_0^3} = 445.5$ inertially while the frame turns at unit rate, so a timing error δt shifts the burn location by $\delta\varphi = (n_{\mathrm{LEO}} - 1)\,\delta t$.

Nominal trajectory and parameters. The nominal free return is found numerically. A scan over Δv and φ locates a free-return family near $\varphi = 4.12$ rad; holding φ fixed, bisection on Δv (`scripts/retarget_nominal.py`) gives $\Delta v = 3.144030$ km/s, for which $\gamma_{\mathrm{EI}} = -6.520°$. Just below this value the return skips past the atmosphere (at $\Delta v = 3.1440$ km/s the vacuum perigee is 147 km), and 0.5 m/s above it the entry is already $-28.7°$. The nominal passes 6,947 km above the lunar surface 3.77 days after TLI and reaches entry interface 7.56 days after TLI at 10.99 km/s, the same as Orion's measured entry speed on Artemis I [4]. Table 2 lists the parameters the investigation varies and the quantities it measures.

Table 2: Investigated parameters and measured outputs.

Parameter	Symbol	Nominal / range	Question
TLI magnitude error	$\delta(\Delta v)$	0; swept over ± 20 mm/s	Q1a
TLI timing error	δt ($\delta\varphi = (n_{\mathrm{LEO}} - 1)\delta t$)	0; swept over ± 2 s	Q1b
Injection error to be corrected	$\delta(\Delta v)$	e.g. ± 1 m/s	Q2
Correction time	t_c	15–20 values, TLI to $\approx$1 day before EI	Q2
Correction magnitude	Δv_c	solved for, along or normal to $\mathbf{v}_E$	Q2

Held fixed: μ, r_0, r_{EI}, nominal Δv and φ, corridor $[-7.4°, -5.25°]$, target $-6.52°$
Measured: γ_{EI} (primary); perilune altitude, time of flight, Δv_c

Events and derived quantities. Let $\mathbf{r}_E = (x + \mu,\ y)$ be the Earth-relative position and

$$\mathbf{v}_E = (v_x - y,\quad v_y + x + \mu) \tag{5}$$

the Earth-relative inertial velocity. *Entry interface* is the first crossing of $|\mathbf{r}_E| = r_{\mathrm{EI}}$ after the lunar flyby, where the entry flight-path angle is

$$\gamma_{\mathrm{EI}} = \arcsin\!\left(\frac{\mathbf{r}_E \cdot \mathbf{v}_E}{|\mathbf{r}_E|\,|\mathbf{v}_E|}\right). \tag{6}$$

A run is flagged as a *lunar impact* if $r_2 < R_M$, and as *no return* if it fails to reach entry interface within 3 time units (≈ 13 days). Along any coasting arc the *Jacobi constant*

$$C = 2\Omega - \left(v_x^2 + v_y^2\right), \qquad \Omega = \tfrac{1}{2}\left(x^2 + y^2\right) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \tag{7}$$

is conserved; it changes only at impulsive burns.

1.5 Why numerical methods are required

The CR3BP is non-integrable: no general closed-form solution exists [6]. The nominal free return, the sensitivity of γ_{EI} to injection errors, and the correction cost all depend on the full nonlinear passage through the Earth–Moon system, including the lunar flyby that drives the error amplification. Patched-conic approximations treat the flyby as an instantaneous switch between two-body problems; they can provide a rough nominal trajectory, but not the sensitivities at the heart of this project. Every answer to Q1 and Q2 therefore comes from numerically integrating Eq. (1). Analytical solutions are used only to verify the implementation (Section 3).

2 Numerical method

2.1 The scheme

We advance Eq. (1) with the classical fourth-order Runge–Kutta method (RK4) at a fixed time step Δt:

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(\mathbf{u}_k), & \mathbf{k}_2 &= \mathbf{f}\left(\mathbf{u}_k + \tfrac{\Delta t}{2}\mathbf{k}_1\right), \\ \mathbf{k}_3 &= \mathbf{f}\left(\mathbf{u}_k + \tfrac{\Delta t}{2}\mathbf{k}_2\right), & \mathbf{k}_4 &= \mathbf{f}\left(\mathbf{u}_k + \Delta t\,\mathbf{k}_3\right), \\ \mathbf{u}_{k+1} &= \mathbf{u}_k + \tfrac{\Delta t}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right). \end{aligned} \tag{8}$$

2.2 Error and stability properties

Truncation error. We show the derivation for a scalar autonomous equation $\dot{u} = f(u)$; for systems the same matching holds term by term with f' the Jacobian. Write $h = \Delta t$ and evaluate f and its derivatives at u_k. Differentiating $\dot{u} = f(u)$ repeatedly ($\ddot{u} = f'f,\ \dddot{u} = f''f^2 + f'^2 f, \ldots$)

and Taylor-expanding the exact solution gives

$$u(t_k+h) = u_k + hf + \frac{h^2}{2}f'f + \frac{h^3}{6}\left(f''f^2 + f'^2 f\right) + \frac{h^4}{24}\left(f'''f^3 + 4f''f'f^2 + f'^3 f\right) + O(h^5). \tag{9}$$

Expanding each stage of Eq. (8) about u_k, with each stage's argument substituted into the next:

$$\begin{aligned}
\mathbf{k}_1 &= f,\\
\mathbf{k}_2 &= f + \tfrac{h}{2}f'f + \tfrac{h^2}{8}f''f^2 + \tfrac{h^3}{48}f'''f^3 + O(h^4),\\
\mathbf{k}_3 &= f + \tfrac{h}{2}f'f + h^2\left(\tfrac{1}{4}f'^2 f + \tfrac{1}{8}f''f^2\right)\\
&\quad + h^3\left(\tfrac{3}{16}f''f'f^2 + \tfrac{1}{48}f'''f^3\right) + O(h^4),\\
\mathbf{k}_4 &= f + hf'f + h^2\left(\tfrac{1}{2}f'^2 f + \tfrac{1}{2}f''f^2\right)\\
&\quad + h^3\left(\tfrac{1}{4}f'^3 f + \tfrac{5}{8}f''f'f^2 + \tfrac{1}{6}f'''f^3\right) + O(h^4).
\end{aligned} \tag{10}$$

Forming $u_k + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$, the coefficients are $\frac{6}{6} = 1$ for hf; $\frac{1}{6}(1+1+1) = \frac{1}{2}$ for $h^2 f'f$; $\frac{1}{6}(\frac{1}{4}+\frac{1}{4}+\frac{1}{2}) = \frac{1}{6}$ for both $h^3 f''f^2$ and $h^3 f'^2 f$; and $\frac{1}{6}(\frac{1}{24}+\frac{1}{24}+\frac{1}{6}) = \frac{1}{24}$, $\frac{1}{6}(\frac{3}{8}+\frac{5}{8}) = \frac{4}{24}$ and $\frac{1}{6}\cdot\frac{1}{4} = \frac{1}{24}$ for the three h^4 terms. Every term of Eq. (9) is matched (we also confirmed this symbolically), so the local truncation error is $O(h^5)$. Over $N = t_f/h$ steps these local errors accumulate, and with $\mathbf{f}$ Lipschitz the global error at a fixed t_f is $N \cdot O(h^5) = O(h^4)$.

Absolute stability. Applying Eq. (8) to the test equation $\dot{u} = \lambda u$ gives $u_{k+1} = R(w)\,u_k$ with $w = \lambda\Delta t$ and

$$R(w) = 1 + w + \frac{w^2}{2} + \frac{w^3}{6} + \frac{w^4}{24}. \tag{11}$$

On the imaginary axis, $w = iy$,

$$|R(iy)|^2 = 1 - \frac{y^6}{72} + \frac{y^8}{576}, \tag{12}$$

so $|R(iy)| \le 1$ if and only if $y^2 \le 8$, i.e. $|y| \le 2\sqrt{2} \approx 2.83$. By contrast, forward Euler gives $|1+iy|^2 = 1 + y^2 > 1$ for every $y \neq 0$, and the stability region of AB2 touches the imaginary axis only at the origin; AB3 and AB4 cover it only up to $|y| \approx 0.72$ and 0.43.

Eigenvalues along the trajectory. Figure 2(a) shows the largest Jacobian eigenvalue magnitude along the nominal free return. It peaks at the low-Earth ends of the trip, where the eigenvalues are $\pm\sqrt{2}\,n$ (real) and $\pm i\,n$, with n the local orbital rate; at the parking orbit $|\lambda|_{\max} = 630$. The negative-real and imaginary eigenvalues must lie inside the stability region. The positive-real one is the physical divergence of neighbouring trajectories, the same mechanism that amplifies TLI errors, and must be resolved accurately rather than damped. At the study time step (Section 3) $|\lambda|\Delta t = 0.019$, far inside the region (Figure 2b).

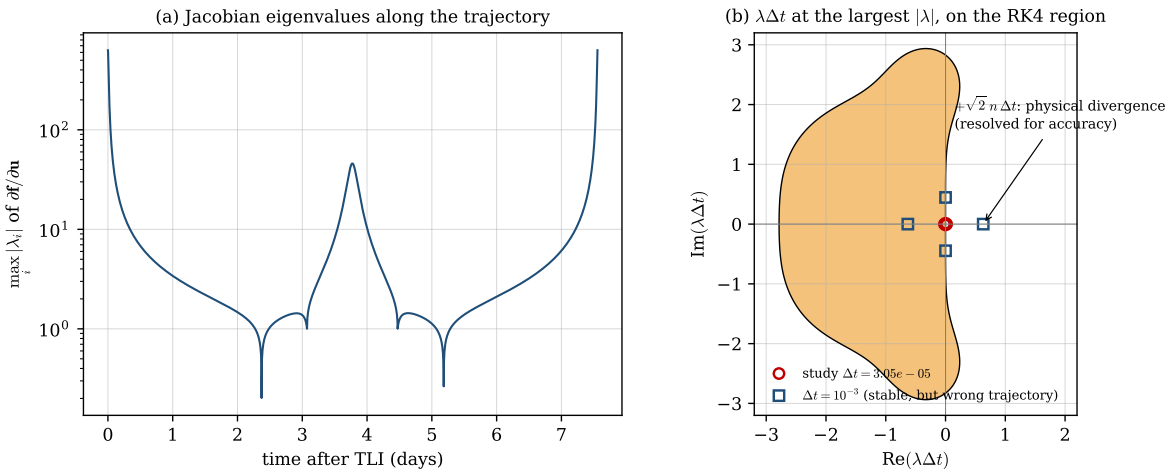


Figure 2: (a) Largest Jacobian eigenvalue magnitude along the nominal free return; the central peak is the lunar flyby. (b) $\lambda\Delta t$ at the state with the largest $|\lambda|$, on the RK4 stability region, for the study time step and for $\Delta t = 10^{-3}$. Even at $\Delta t = 10^{-3}$ every decaying and oscillatory mode is stable, yet that run never returns to Earth: the step is limited by accuracy, not stability.

2.3 Justification: accuracy, stability, and cost

- **Accuracy.** The required γ_{EI} accuracy is a small fraction of a 2.15° corridor, and the sensitivities in Q1 must be resolved at the sub-mm/s level. A fourth-order method reaches this accuracy with orders of magnitude fewer steps than a first- or second-order method.

- **Stability.** Near Earth the motion has oscillatory modes $\pm i\, n$ on the imaginary axis. RK4's stability region contains the segment $|\lambda\Delta t| \leq 2\sqrt{2}$ of that axis; forward Euler's and AB2's contain none of it, so both would artificially amplify orbital oscillations at any Δt.

- **Not stiff, so explicit is appropriate.** The Jacobian eigenvalues are largest near Earth, but there the solution itself varies on the same short timescale. The step is limited by *accuracy*, not stability, so an implicit method (backward Euler, trapezoid, BDF2) would add a nonlinear solve every step with no benefit. Figure 2(b) shows this directly: at $\Delta t = 10^{-3}$, $|\lambda\Delta t| = 0.63$ for the real modes and 0.45 for the oscillatory ones, well inside the region, yet the trajectory never returns to Earth.

- **One step, not multistep.** A midcourse correction changes the velocity instantly, and locating entry interface needs a partial step. A one-step method handles both directly; a multistep method such as AB4 would have to restart its history after every burn. (At our time step $|\lambda|\Delta t \approx 0.02$ is also inside AB4's region, so stability alone would not rule AB4 out.)

- **Cost, and why adaptive stepping is not used.** The problem is multi-scale: the parking-orbit period is 0.0141 time units against a trip of 1.740. Adaptive step-size control would therefore reduce function evaluations. However, fixed-step RK4 at the study time step $\Delta t = 3.05 \times 10^{-5}$ (11.4 s) takes about 2 s per run in plain Python, and the full study requires a few

hundred runs. Adaptive control would add an error estimator, step-rejection logic, and a less transparent convergence study, without changing the answers.

2.4 Algorithm: advancing from t_k to t_{k+1}

1. Evaluate $\mathbf{k}_1, \ldots, \mathbf{k}_4$ from $\mathbf{u}_k$ and form $\mathbf{u}_{k+1}$ via Eq. (8).
2. **Entry-interface event.** If, after the flyby, $|\mathbf{r}_E| - r_{\mathrm{EI}}$ changes sign between $\mathbf{u}_k$ and $\mathbf{u}_{k+1}$, find $\tau \in (0, \Delta t)$ by root-finding (Brent's method) such that a single RK4 step of length τ from $\mathbf{u}_k$ lands exactly on $|\mathbf{r}_E| = r_{\mathrm{EI}}$, then evaluate Eq. (6). Locating the event with an RK4 step preserves $O(\Delta t^4)$ accuracy; taking the minimum over sampled steps would introduce an $O(\Delta t^2)$ error.
3. **Lunar impact.** If $r_2 < R_M$, flag the run and stop.
4. **Midcourse correction.** At the step k with t_k nearest t_c, add Δv_c along $\mathbf{v}_E/|\mathbf{v}_E|$ (or along the in-plane normal to it) to the velocity components of $\mathbf{u}_k$ before stepping. The position is unchanged, so the same vector is added to the rotating-frame velocity.

2.5 Targeting

Retargeting the nominal (find Δv such that $\gamma_{\mathrm{EI}} = -6.52°$) and sizing corrections (find Δv_c such that $\gamma_{\mathrm{EI}} = -6.52°$) are scalar root-finding problems wrapped around our RK4 integrator. We use bisection or `scipy.optimize.brentq`; root-finders are permitted, external time integrators are not.

3 Implementation and verification

3.1 Verification against independent results

Table 3: Verification tests.

Test	What it checks	Pass criterion
Two-body exact solution ($\mu = 0$)	Integrator and rotating-frame terms coded correctly	Error vs. exact solution decreases as $O(\Delta t^4)$
Published periodic orbit (JPL)	Full three-body equations, including every Moon term	Closure error after one period decreases as $O(\Delta t^4)$
Jacobi constant	Conservation law of the coasting CR3BP	$\max_t \lvert C(t) - C(0)\rvert$ decreases as $O(\Delta t^4)$
Convergence rate	Method achieves its theoretical order	Observed order ≈ 4 in the asymptotic range
Event-location order	γ_{EI} inherits fourth-order accuracy	γ_{EI} error decreases as $O(\Delta t^4)$

Two-body exact solution. Setting $\mu = 0$ in Eq. (1) places Earth at the origin with unit gravitational parameter, so the system reduces to Kepler motion viewed from a frame rotating at unit rate, for which the inertial solution is known exactly. We choose an ellipse with perigee radius $r_p = r_0$ and eccentricity $e = 0.97$, close to the shape of the translunar ellipse, giving

$$a = \frac{r_p}{1-e} = 0.5691, \qquad b = a\sqrt{1-e^2}, \qquad n = a^{-3/2}, \qquad P = \frac{2\pi}{n} = 2.698. \tag{13}$$

Measuring time from perigee, the eccentric anomaly E satisfies Kepler's equation $nt = E - e\sin E$, which we solve with Newton's method,

$$E_{j+1} = E_j - \frac{E_j - e\sin E_j - nt}{1 - e\cos E_j}, \qquad E_0 = \pi, \tag{14}$$

to $|E_{j+1} - E_j| < 10^{-15}$. The inertial state follows from

$$\mathbf{r}_{\text{in}} = \begin{bmatrix} a(\cos E - e) \\ b\sin E \end{bmatrix}, \qquad \mathbf{v}_{\text{in}} = \frac{n}{1 - e\cos E}\begin{bmatrix} -a\sin E \\ b\cos E \end{bmatrix}, \tag{15}$$

and is rotated into the CR3BP frame by

$$\mathbf{r}_{\text{rot}} = \mathsf{R}(-t)\,\mathbf{r}_{\text{in}}, \qquad \mathbf{v}_{\text{rot}} = \mathsf{R}(-t)\,\mathbf{v}_{\text{in}} + \begin{bmatrix} y_{\text{rot}} \\ -x_{\text{rot}} \end{bmatrix}, \qquad \mathsf{R}(-t) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}, \tag{16}$$

where the second term in $\mathbf{v}_{\text{rot}}$ comes from differentiating the time-dependent rotation. The integration starts from the exact perigee state $\mathbf{u}_0 = [\,r_p,\ 0,\ 0,\ v_p - r_p\,]^T$ with $v_p = \sqrt{(1+e)/r_p} = 10.74$ (11.0 km/s), the same speed as TLI and lunar-return entry, and calls the same right-hand-side function used for all results with $\mu = 0$. We measure $\varepsilon(\Delta t) = \|\mathbf{r}_{\text{num}}(t_f) - \mathbf{r}_{\text{exact}}(t_f)\|_2$ at $t_f = P/2$ (apogee) and $t_f = P$ (back at perigee), with $\Delta t = t_f/N$ so that the march lands exactly on t_f. Figure 3 shows observed orders 4.02–4.21. At every Δt the full-orbit error is about 36× the half-orbit error, because most of the error accumulates during the perigee passage; this is why perigee sets the time step.

Published periodic orbits. Because $\mu = 0$ multiplies every Moon term by zero, the two-body test cannot detect a mistake in those terms. We therefore also integrate three planar orbits from JPL's Earth–Moon CR3BP catalog [5], each with the catalog's own $\mu = 0.01215058561$: a distant retrograde orbit of Artemis I's size (period 14.0 days), and L1 and L2 Lyapunov orbits (periods 13.3 and 15.3 days), which are strongly unstable (stability indices of about 690 and 530), so any error in the Coriolis or gravity terms is amplified hundreds of times per period. Each must return to its initial state after exactly one published period. For all three the Jacobi constant computed from the published state agrees with the catalog value to 14 digits, and the closure error falls at observed orders 3.87–4.10 (Figure 4). An error in $\mathbf{f}$ would make the closure error level off at a nonzero value instead.

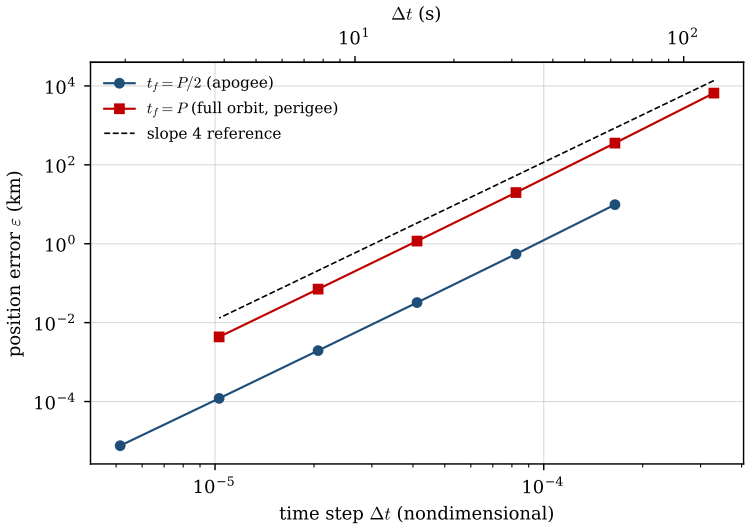


Figure 3: RK4 position error against the exact two-body solution ($\mu = 0$, $e = 0.97$, perigee at r_0) after half an orbit and after a full orbit. Both converge at fourth order.

Jacobi constant. Along the nominal free return, from TLI to entry interface, the maximum drift $\max_k |C(t_k) - C(0)|$ falls at observed orders 4.03–4.22 (Figure 4), reaching 3.5×10^{-8} at the study time step.

Convergence study. The self-convergence study in `FreeReturnTrajectory.ipynb` integrates the nominal to $t_f = 1.7$ (just before entry interface) and compares the full state against a reference with $\Delta t = t_f/2^{17}$, for $\Delta t = t_f/2^{12}$ to $t_f/2^{16}$. The observed orders are 4.50, 4.21, 4.12, 4.14. At $\Delta t \approx 10^{-3}$ the trajectory never returns to Earth, so that step lies outside the asymptotic range.

3.2 Selecting simulation parameters

The time step must make the numerical error in γ_{EI} negligible compared with both (i) the corridor width, 2.15°, and (ii) the change in γ_{EI} produced by the smallest injection error we resolve, $\approx$ 0.1 mm/s. At the nominal $d\gamma_{\mathrm{EI}}/d(\Delta v) = -0.14°$ per mm/s, so requirement (ii) is 0.014° and requirement (i), 1% of the corridor, is 0.0215°. Figure 5 plots the γ_{EI} error against Δt, measured against a reference with $\Delta t = 1.9 \times 10^{-6}$. The error falls at observed orders 4.39–4.03, which also confirms that the event location keeps fourth-order accuracy. We use $\Delta t = 3.05 \times 10^{-5}$ (11.4 s), where the γ_{EI} error is 2.3×10^{-4} degrees, about 60× below the tighter requirement, at about 2 s per run. This Δt is used for all results in Section 4.

4 Results

All results use the nominal of Section 1 ($\Delta v = 3.144030$ km/s, $\varphi = 4.12$ rad, $\gamma_{\mathrm{EI}} = -6.520°$; Figure 1a) and the study time step $\Delta t = 3.05 \times 10^{-5}$ selected in Section 3. They are produced by

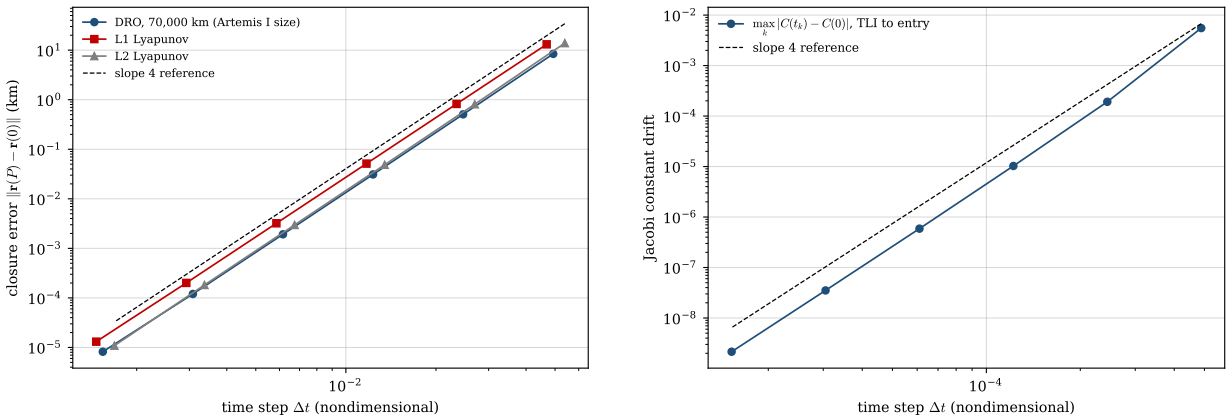


Figure 4: Left: closure error of three JPL catalog orbits after one published period. Right: Jacobi-constant drift along the nominal free return. Both decrease at fourth order.

`scripts/study_q1.py` and `scripts/study_q2.py`.

4.1 Q1: How accurate must the TLI burn be?

Procedure. For Q1a we vary the TLI magnitude by $\delta(\Delta v) \in [-20, 20]$ mm/s with φ fixed; for Q1b we hold Δv and start the burn $\delta t \in [-2, 2]$ s late, so $\delta\varphi = (n_{\mathrm{LEO}} - 1)\,\delta t$. Each sweep uses 81 runs. The corridor edges are then located by Brent's method on $\gamma_{\mathrm{EI}}(\delta) = -5.25°$ and $\gamma_{\mathrm{EI}}(\delta) = -7.4°$, with runs that skip out of the atmosphere counted as $\gamma_{\mathrm{EI}} = 0$.

Results. Figure 6 shows that a larger or later burn steepens the entry. Near the nominal the sensitivities are

$$\frac{\partial \gamma_{\mathrm{EI}}}{\partial(\Delta v)} = -0.1405° \text{ per mm/s}, \qquad \frac{\partial \gamma_{\mathrm{EI}}}{\partial t_{\mathrm{burn}}} = -5.71° \text{ per s}. \tag{17}$$

A 20 mm/s error, about six parts per million of the burn, already moves γ_{EI} from $-6.52°$ to $-2.4°$ or $-8.9°$. The response to timing is stronger and more nonlinear: a burn 0.6 s early returns with a vacuum perigee above entry interface and skips out, while a burn 2 s late enters at $-13.5°$. Table 4 gives the largest errors that keep γ_{EI} in the corridor.

Table 4: Largest TLI errors that keep the entry inside the corridor.

Error	Shallow edge ($-5.25°$)	Steep edge ($-7.4°$)	Total window
Magnitude $\delta(\Delta v)$	-8.14 mm/s	$+6.69$ mm/s	14.8 mm/s (4.7 ppm of Δv)
Timing δt	-0.200 s	$+0.165$ s	0.365 s

Interpretation. Both windows are asymmetric, wider on the shallow (under-burn, early) side, because γ_{EI} curves toward 0° as the vacuum perigee rises toward the entry interface. The answer to Q1 is that an open-loop TLI must be accurate to within $-8.1/+6.7$ mm/s in magnitude and $-0.20/+0.17$ s in timing, millimetres per second and fractions of a second as we hypothesized.

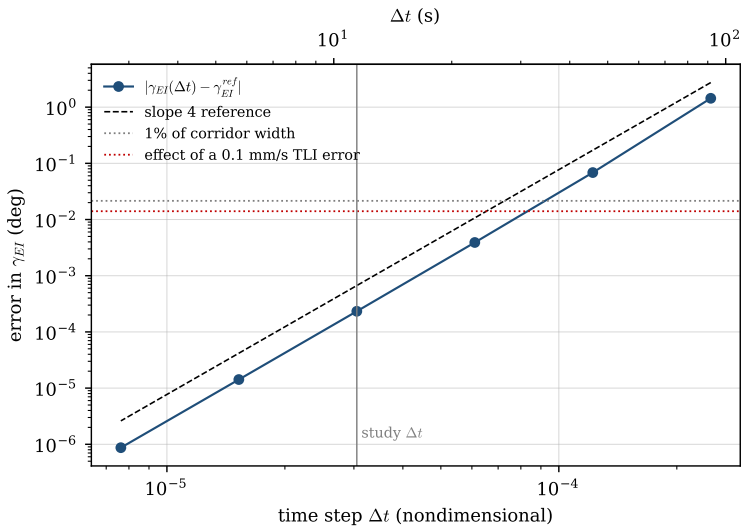


Figure 5: Error in the entry flight-path angle against time step, with the two accuracy requirements. The vertical line marks the study time step.

That is far tighter than an open-loop burn of more than 3 km/s can be expected to deliver, so a free return is therefore only abort-safe in practice if it is followed by midcourse corrections, which motivates Q2.

4.2 Q2: What does correcting an injection error cost?

Procedure. We apply injection errors of $\delta(\Delta v) = +1$ m/s, which enters at $-39.1°$ uncorrected, and -1 m/s, which never returns. Both are far outside the corridor. Each erroneous trajectory is marched once and stored. At 18 correction times t_c from 0.1 to 6.5 days after TLI, the run restarts from the stored state, applies an impulsive burn Δv_c, and continues to entry interface. Δv_c is the smallest-magnitude root of $\gamma_{\mathrm{EI}}(\Delta v_c) = -6.52°$, found by expanding a bracket from zero and refining with Brent's method. The plan in Table 2 fixes the burn along $\mathbf{v}_E$. Because that direction turns out to be a poor choice after the flyby, we also solve for an in-plane burn *normal* to $\mathbf{v}_E$, and we take the cheaper of the two as an upper bound on the optimal single-burn cost.

Results. Figure 7 and Table 5 give the results. Until the spacecraft nears the Moon ($t_c \leq 3.1$ days) the along-velocity burn is always the cheaper one, and its cost rises steadily from 2.65 m/s at $t_c = 0.1$ days to about 40 m/s at 3.5 days. After the flyby the along-velocity burn becomes very expensive (200–470 m/s from 4.2 days on), while the normal burn costs 38–65 m/s for the $+1$ m/s error and 53–163 m/s for -1 m/s, growing as the spacecraft falls back toward Earth.

Interpretation. Before the flyby, the correction is essentially an energy correction. A burn changes the specific energy by $|\mathbf{v}_E|\,\Delta v_c$, and the product $|\mathbf{v}_E|\,|\Delta v_c|$ stays between 10.6 and 11.8 (in km/s times m/s) for the first two days. That matches the energy error the TLI mistake introduced, 10.9×1 in the same units. The correction grows simply because the spacecraft slows down as it

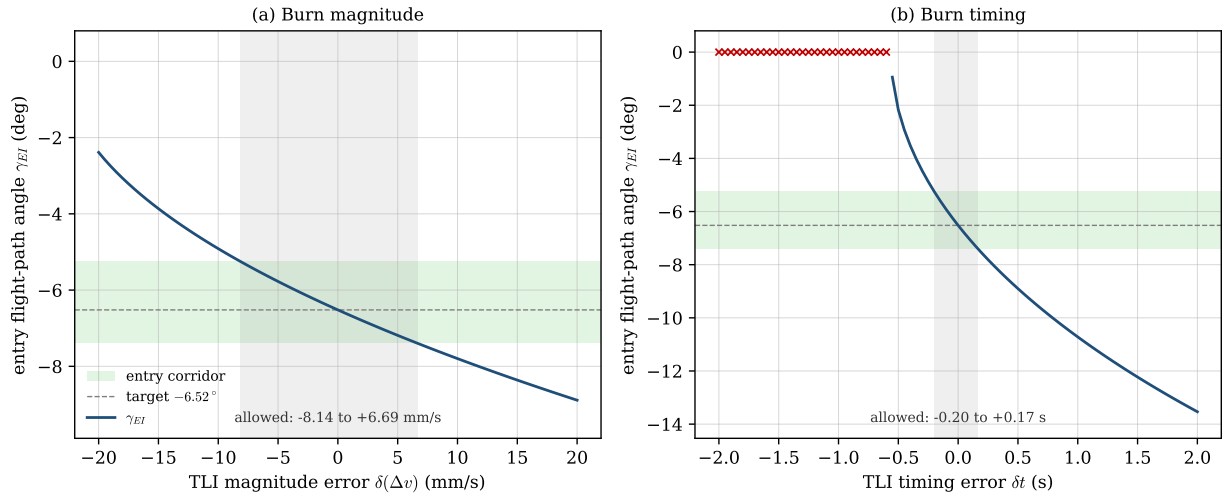


Figure 6: Entry flight-path angle against (a) TLI magnitude error and (b) TLI timing error. Green: entry corridor $[-7.4°, -5.25°]$; grey: the range of errors that stays inside it. Crosses mark early burns whose return misses the atmosphere entirely.

Table 5: Cheapest single correction (of the two directions) restoring $\gamma_{\mathrm{EI}} = -6.52°$, for a 1 m/s TLI error.

t_c (days after TLI)	0.10	1.23	2.36	3.49	3.86 (after flyby)	4.99	6.50
TLI +1 m/s: $\lvert\Delta v_c\rvert$ (m/s)	2.6	8.5	16.0	32.5	37.5	44.4	64.7
TLI −1 m/s: $\lvert\Delta v_c\rvert$ (m/s)	2.7	8.5	16.1	40.0	53.0	86.2	163.3

climbs (from 11 km/s at TLI to 4.0 km/s at 0.1 days and 0.9 km/s at 2 days). This is the Oberth effect working in reverse. After the flyby the energy relative to Earth is no longer what sets the return: the error has become a displacement of the return perigee, and changing the perigee from far out takes a burn that rotates the velocity, so the normal direction wins and the cost keeps rising as the lever arm shrinks.

The answer to Q2 is therefore that correcting *early* is cheapest by a wide margin. The same 1 m/s error costs 2.7 m/s to correct 2.4 hours after TLI, 14–20 times more just after the flyby (38–53 m/s), and 24–62 times more one day before entry (65–163 m/s). This confirms our hypothesis that the cost depends strongly on t_c. The flyby does mark a transition, in the best burn direction and in the asymmetry between over- and under-burns, but the cost is already climbing steeply on the way out, not only after the Moon.

4.3 Scope and limitations

The corridor is used as a targeting criterion; atmospheric flight, heating, skip dynamics, and entry guidance are outside the model. The planar, circular CR3BP neglects the Moon's inclination and eccentricity, solar gravity, and J_2, so absolute values will differ from a real mission; the orders of

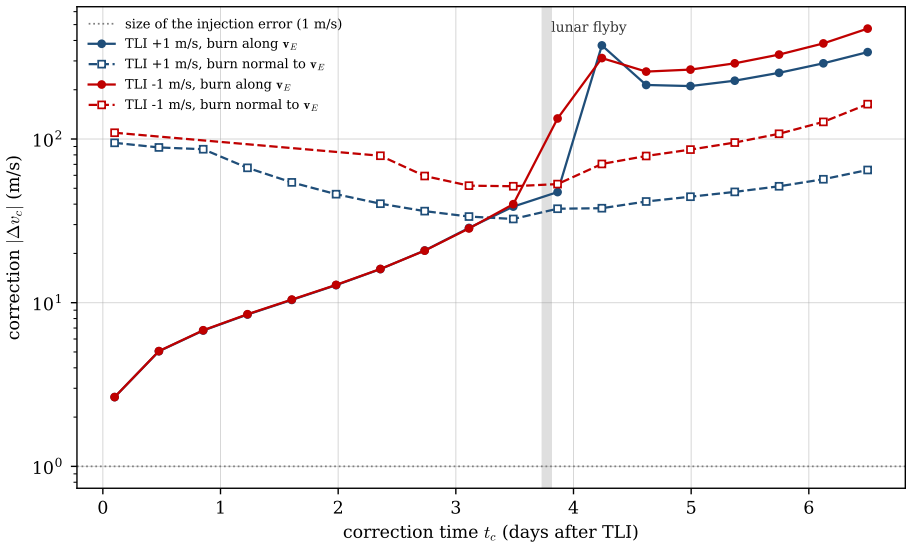


Figure 7: Midcourse correction needed to restore $\gamma_{\text{EI}} = -6.52°$ after a ± 1 m/s TLI error, against the time of the correction, for burns along and normal to the Earth-relative velocity $\mathbf{v}_E$. The grey band is the lunar flyby (3.77 days).

magnitude of the sensitivities and the trends in correction cost are the transferable results. Burns are impulsive, magnitude and timing errors are studied separately, and Q2 compares two burn directions rather than optimizing over all of them, so its costs are upper bounds on the best single burn.

As a check on how far the absolute numbers transfer, the notebooks `ApolloComparison.ipynb` and `ArtemisIIComparison.ipynb` run the same model from the flown post-TLI states of Apollo 8 and Artemis II (JPL Horizons ephemeris). Once its lunar pass is matched to the real one with a velocity adjustment of 8–14 m/s, and with the Moon placed at its actual distance for that date, the model reproduces Artemis II's maximum distance from Earth, entry time and entry speed to within 1%. Its entry angle, however, is only reliable when started within the final ∼20 hours, because the real flyby tilted the return leg about 20° out of the Moon's orbital plane. This is the same extreme sensitivity of γ_{EI} that Q1 quantifies, and it is why real missions plan several midcourse corrections.

5 Contributions

Every member was actively involved in all parts of the project: choosing the premise, selecting and justifying the method, implementing and verifying the code, producing the results, and writing the report. Decisions were made together, and each member reviewed the others' work. Each of us also led particular areas, as described below. We estimate our contributions as equal, at 25% each.

Zohair Hasson (25%) came up with the final prompt the project is built around: how accurate a

free-return translunar burn must be, and what correcting an injection error costs. He led the technical writing of the report and shaped how the premise, method and findings are presented. He also modified the code and results so that the studies addressed the research questions directly.

Jammir Aloya (25%) did most of the coding, including the core numerical implementation that every study relies on. He made sure each method in the code was valid and consistent with the report, so the equations, assumptions and parameters in the document match what the code computes. He also carried out most of the calculations behind the verification and results.

Akhil Boyapati (25%) created and maintained the GitHub repository and organized it so the numerical method is kept separate from the studies that use it. He revised the code for correctness and clarity, and worked on both the implementation and the numerical method, including its justification and verification.

Jeehyung Yoo (25%) helped produce the results and interpret what they mean for the research questions. He checked the validity of all the project documents, comparing the report against the code and the results so that every claim, number and figure is consistent and correct.

6 Reproducibility

All code is available at https://github.com/boyakhil978/FreeReturnTrajectory. The repository separates the numerical method from the studies that use it:

- `freereturn/integrators.py`: the numerical method, `rk4_step` and the fixed-step marcher `u, times = rk4(f, u_start, t_end, dt)`;
- `freereturn/dynamics.py`: right-hand side $\mathbf{f}(\mathbf{u})$, its Jacobian, and the Jacobi constant C;
- `freereturn/trajectory.py`, `freereturn/entry.py`: TLI initial state, γ_{EI}, impulsive burns, and the marcher from TLI to entry interface with event location;
- `scripts/`: targeting of the nominal (`retarget_nominal.py`), the verification and time-step selection studies (`verify_*.py`, `select_dt.py`, `stiffness_along_trajectory.py`), and the results (`study_q1.py`, `study_q2.py`); each writes its figure to `figures/` and its numbers to `results/`;
- `ApolloComparison.ipynb`, `ArtemisIIComparison.ipynb`: comparison of the model with Apollo 8 and Artemis II flight data (JPL Horizons, cached in `data/`);
- `tests/`: automated checks of the model code (`python -m pytest`);
- `README.md`: instructions to reproduce every figure.

Acknowledgments

Generative AI (Anthropic's Claude, used through Claude Code and the Claude chat interface) was used throughout the project, and every result was checked by the team.

- **Ideation and premise.** A set of motivating free body diagrams were selected that were interesting to us and AI was used to analyze and help us narrow down our choices.
- **Method selection and justification.** Suggesting the triangular point L_4 as an equilibrium at which to evaluate the Jacobian, and using eigenvalue magnitudes for the stiffness ratio (both credited in `FreeReturnTrajectory.ipynb`); checking the RK4 truncation-error derivation symbolically.
- **Implementation and testing.** Organizing the code into the `freereturn` package with automated tests; writing the event location, verification, time-step selection and Q1/Q2 study scripts; retrieving the JPL periodic-orbit and Horizons data and writing the Apollo 8 and Artemis II comparison notebooks. The RK4 integrator and equations of motion were written by the team, and AI-written code was reviewed and run by the team.
- **Figures and results.** Generating the figures from the study scripts and drafting and editing report text, including the results discussion; all numbers in the report come from the repository's code.

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